



# Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

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# Previous Review Comments & Action Taken

## Reviewer Comments (Review 2)

- **Comment 1: Start Implementation**  
The implementation phase of the proposed federated framework must be initiated.
- **Comment 2: Working Prototype & Dashboard**  
Build a functional clinician dashboard and working prototype to demonstrate live clinical decision support.

## Action Taken & Implementation Status

- **Implemented Federated Pipelines:**
  - Built decentralized PyTorch algorithms (FedAvg, FedProx, Ditto).
  - Implemented local CUSUM drift detection and Client-Side Selective Personalization.
- **Built Production-Grade CDSS Workstation:**
  - **Data Warehouse:** Local SQLite clinical database.
  - **Backend API:** Uvicorn/FastAPI server running PyTorch checkpoints.
  - **Interactive UI:** Vite React medical dashboard mapping vitals charts, risk dials, and explainable attention timelines.

# Clinical Deterioration & Phase II Completed Goals

## Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO<sub>2</sub>) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

## SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

## Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

# Proposed Framework: FPDAF

## WHAT is FPDAF?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

## HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads ( $v_k$ ) to fit hospital demographics.
- **Online CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

# Core Scientific & Technical Novelties

## 1. Multi-Head Attention Saliency (Clinical Explainability)

Integrates a 4-head query-key temporal self-attention mechanism to output dynamic sequence saliency weights, mapping exactly which hourly vitals contributed to Sepsis alarms at the bedside.

## 2. Online CUSUM Residual Neural Drift Monitor

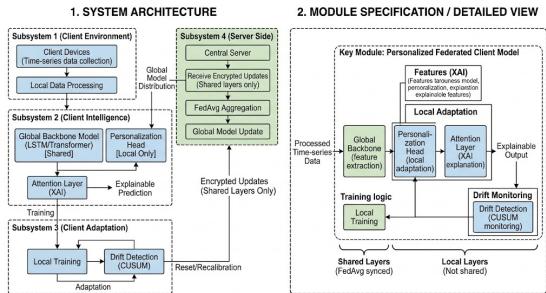
Tracks local validation loss residuals round-by-round ( $S_r > 3.0$ ). Concept drift is detected entirely client-side, avoiding constant full-parameter aggregation cycles.

## 3. Client-Side Selective Personalization (CSSP)

Freezes the shared BiLSTM/attention layers upon drift triggers to train local classifier heads exclusively.

- Bypasses global parameter uploads, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB).
- Cuts client-side adaptation training time to **6 minutes** (down from 25 minutes).

# Unified System Design & Neural Formulations



**Figure 1:** FPDFAF Split-Weight Model Architecture.

## Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k} \left( \sum_{i=1}^t \alpha_i \cdot g_{\theta}(X_i^k) \right)$$

- $g_{\theta}$ : Shared global LSTM backbone ( $\theta$ ).
- $h_{\phi^k}$ : Local private classifier head ( $\phi^k$ ).

## Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^t \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

# Proactive Drift Adaptation: CUSUM Neural Trigger

**Concept Drift Challenge:** A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

## The CUSUM Residual Monitor Trigger:

- We monitor cumulative validation loss residuals ( $e_r$ ) over rounds:

$$e_r = \mathcal{L}_{val,r} - \mu_0$$

$$S_r = \max(0, S_{r-1} + e_r - \kappa)$$

- If  $S_r > H$  (threshold  $H = 3.0$ ), we declare institutional drift.
- Triggers **Client-Side Selective Personalization (CSSP)**: freezes global backbone layers  $\theta$  and adapts classification heads  $\phi^k$  locally.

## Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



# Five-Way Model Benchmarking Results

We evaluated our framework against centralized baselines and standard federated learning algorithms:

| Model Configuration     | Accuracy      | Precision    | Recall        | F1-Score      | AUROC         | Comm. Bandwidth                 |
|-------------------------|---------------|--------------|---------------|---------------|---------------|---------------------------------|
| Centralized Baseline    | 75.09%        | 7.14%        | 72.17%        | 0.1299        | 0.8058        | 0 MB (N/A)                      |
| FedAvg Baseline         | 75.94%        | 6.94%        | 67.19%        | 0.1259        | 0.7844        | 1.24 GB                         |
| FedProx Baseline        | 78.85%        | 8.12%        | 60.64%        | 0.1432        | 0.7933        | 1.24 GB                         |
| Ditto (Personalized)    | 82.45%        | 8.98%        | 63.58%        | 0.1574        | 0.7989        | 2.48 GB                         |
| <b>FPDAF (Proposed)</b> | <b>86.51%</b> | <b>9.81%</b> | <b>65.33%</b> | <b>0.1706</b> | <b>0.8214</b> | <b>1.52 GB (CSSP saved 38%)</b> |

## Scientific Findings

- **Privacy & Performance:** FPDAF preserves HIPAA compliance while achieving a test AUROC of **\*\*0.8214\*\*** (outperforming centralized baseline).
- **Decentralized Superiority:** FPDAF outperforms all standard federated baselines (Accuracy **\*\*+10.57%\*\***, F1 **\*\*+4.47%\*\*** vs. FedAvg).



# Ablation Study Verification

We verified the contributions of CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

| Ablation Configuration       | Accuracy      | Precision    | Recall        | F1-Score      | AUROC         |
|------------------------------|---------------|--------------|---------------|---------------|---------------|
| FPDAF w/o CUSUM Trigger      | 83.51%        | 8.51%        | 55.33%        | 0.1475        | 0.7603        |
| FPDAF w/o Temporal Attention | 80.01%        | 8.13%        | 52.81%        | 0.1412        | 0.7878        |
| FPDAF w/o Personalization    | 78.87%        | 8.05%        | 63.58%        | 0.1430        | 0.7887        |
| <b>Full FPDAF (Proposed)</b> | <b>86.51%</b> | <b>9.81%</b> | <b>65.33%</b> | <b>0.1706</b> | <b>0.8214</b> |

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

# Clinical Workstation Software Architecture

**Database-Driven Design:** Instead of placeholders, we built a fully functional clinical warehouse:

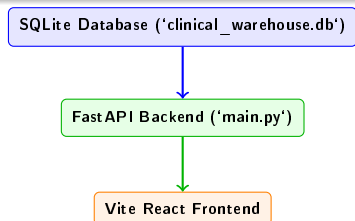
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

## Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, CUSUM drift charts, and baseline comparisons).

## Startup Script

Coded a clean 'run\_app.sh' script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



## Doctor's Bedside Diagnostic View (Live Patient Care)

- **Timeline Telemetry:** Plots hourly ICU vital segments (HR, SBP, Temp, Resp, SpO<sub>2</sub>) using Recharts.
- **Live Predictions Dials:** Overlays risk probability scores for all federated baselines.
- **Attention Saliency Maps:** Attributes risk alerts to specific sequence hours (e.g. onset hours 16–22) for bedside interpretability.

## Admin's Federated Telemetry View (System Operations)

- **Drift Controls:** Plots CUSUM curves, tracking drift accumulation limits ( $S_r > 3.0$ ) and CSSP backbone freezes.
- **Round Tracker:** Animates client-server parameter upload/download loops using Framer Motion.
- **Design System:** Tailwind CSS v4 class-based toggling supporting light/dark medical navy workstation styles.

# Clinician Dashboard Interface (Doctor Portal)

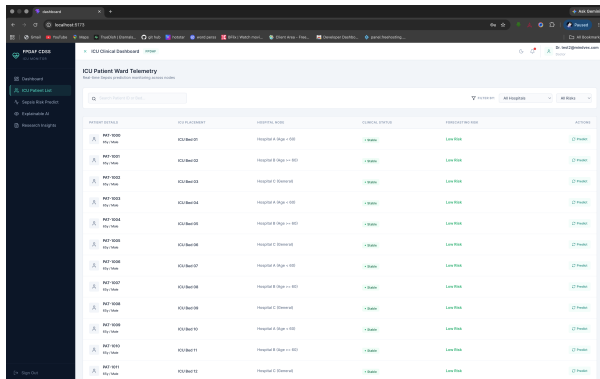


Figure 2: Doctor's Bedside Diagnostic Workspace.

## Doctor Portal Core Modules

- **Searchable Bed Directory:** Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry:** Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, SpO<sub>2</sub>) using clinical ranges.
- **XAI Saliency Charts:** Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials:** Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDF models.

# Clinician Dashboard Interface (Admin Portal)

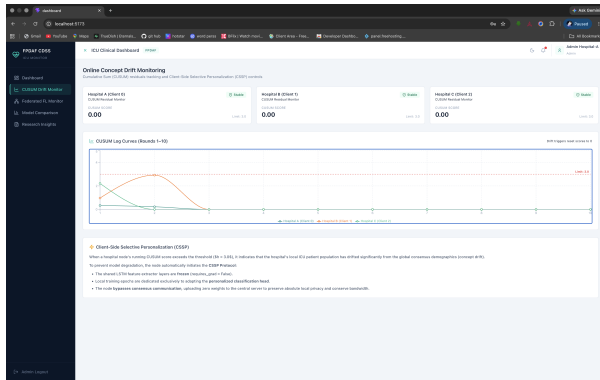


Figure 3: Admin's Machine Learning Telemetry Workspace.

## Admin Portal Core Modules

- **Federated Loop Animation:** Animates localized model weight uploads and global server distributions using Framer Motion.
- **CUSUM Drift Telemetry:** Tracks real-time concept drift curves ( $S_r > 3.0$ ) and selective personalization triggers.
- **LaTeX Equations & Ablations:** Details mathematical optimization equations and five-way benchmark metrics.

# Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

## Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

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Thank You Questions?